COURSE 6 · BIOLOGY

Biology

Printable flashcards

169 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 6.1 · The chemistry of life

EQUATION Write the equation: Condensation (making a polymer)	$\text{monomer} + \text{monomer} \rightarrow \text{dimer} + \text{H}_2\text{O}$
EQUATION Write the equation: Enzyme kinetics (Michaelis–Menten)	$v = \frac{V_{\max}[S]}{K_m + [S]}$
EQUATION Write the equation: Enzyme reaction scheme	$E + S \rightleftharpoons ES \rightarrow E + P$
KEY IDEA Complete the key idea: Water's polarity and hydrogen …	Water's polarity and hydrogen bonding make it the medium of life
KEY IDEA Complete the key idea: Carbohydrates, lipids, …	Carbohydrates, lipids, proteins, nucleic acids
KEY IDEA Complete the key idea: Polymers form by …	Polymers form by condensation and break by hydrolysis
KEY IDEA Complete the key idea: Enzymes lower activation energy; …	Enzymes lower activation energy; they denature outside their optimum
QUICK CHECK Enzymes are mostly	proteins
QUICK CHECK Water dissolves many substances because it is	polar
QUICK CHECK DNA and RNA are	nucleic acids

QUICK CHECK Human enzyme optimum temperature is about	37 °C
PROBLEM Proteins are polymers of	amino acids. Twenty kinds of amino acid.
PROBLEM Which molecule forms cell membranes?	phospholipid. Lipids with a water-loving head and water-avoiding tails.
PROBLEM Why does an enzyme stop working at high temperature?	its shape changes and the active site no longer fits. Denaturation.
PROBLEM Why does a lake freeze from the top down, protecting fish?	ice is less dense than water and floats. Hydrogen bonding gives ice an open structure.
PROBLEM Using $v = V_{max}[S]/(K_m + [S])$, find v when $[S] = K_m$ and $V_{max} = 10$ units.	5 units. $v = 10 \times K_m/(2K_m) = 5$: half the maximum.
PROBLEM Breaking starch into glucose during digestion is	hydrolysis. Water is used to split the bonds.
PROBLEM At very high substrate concentration, enzyme rate levels off because	all active sites are occupied. Saturation.
PROBLEM Explain how an enzyme's specificity follows from its structure.	An enzyme is a protein chain folded into a precise three-dimensional shape. The active site is a pocket whose shape and pattern of charges match particular substrate molecules. Only substrates that fit and interact correctly can bind and react, so each enzyme catalyses one reaction or a small family of reactions.

Module 6.2 · Cells

EQUATION Write the equation: Surface-to-volume ratio of a cube of side L	$\frac{SA}{V} = \frac{6L^2}{L^3} = \frac{6}{L}$
EQUATION Write the equation: Magnification	$M = \frac{\text{image size}}{\text{actual size}}$
EQUATION Write the equation: Fick's law (diffusion rate)	$\text{rate} \propto \frac{A\,\Delta c}{d}$
KEY IDEA Complete the key idea: All living things are …	All living things are made of cells; cells come from cells
KEY IDEA Complete the key idea: Prokaryotes: no nucleus. …	Prokaryotes: no nucleus. Eukaryotes: nucleus and organelles
KEY IDEA Complete the key idea: Diffusion and osmosis are …	Diffusion and osmosis are passive; active transport uses energy
KEY IDEA Complete the key idea: Surface-to-volume ratio …	Surface-to-volume ratio limits cell size
QUICK CHECK Cell theory says cells come from	other cells
QUICK CHECK Plant cells have, but animal cells lack,	a cell wall
QUICK CHECK Osmosis is the movement of	water
QUICK CHECK As a cell grows, its SA:V ratio	decreases
PROBLEM Which cells lack a nucleus?	bacteria. Prokaryotes have no nucleus.
PROBLEM Where in the cell is aerobic respiration carried out?	mitochondrion. The powerhouse.

PROBLEM A cell 20 µm across appears 8 mm across in a micrograph. What is the magnification?	400 ×. 8000 ÷ 20 = 400.
PROBLEM What is the surface-area-to-volume ratio of a cube-shaped cell with sides 2 µm? (μm^{-1})	3. $6/L = 6/2 = 3\ \mu m^{-1}$.
PROBLEM And for sides of 10 µm?	0.6. $6/10 = 0.6\ \mu m^{-1}$: five times smaller.
PROBLEM A red blood cell is placed in pure water. What happens?	water enters by osmosis and it swells, possibly bursting. Water moves towards the higher solute concentration inside the cell.
PROBLEM Which process needs energy from the cell?	active transport. Moving substances against a gradient.
PROBLEM What evidence suggests mitochondria were once free-living bacteria?	Mitochondria have their own circular DNA, like bacteria; they have their own bacteria-like ribosomes; they divide by splitting, as bacteria do; and they are surrounded by a double membrane, consistent with being engulfed. Gene sequences show their closest relatives are a group of bacteria.

Module 6.3 · Metabolism

EQUATION Write the equation: ATP hydrolysis	$\mathrm{ATP + H_2O \rightarrow ADP + P_i + energy}$
EQUATION Write the equation: Photosynthesis	$\mathrm{6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2}$
EQUATION Write the equation: Aerobic respiration	$\mathrm{C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O\ (\approx 30\ ATP)}$
EQUATION Write the equation: Fermentation in yeast	$\mathrm{C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2}$
KEY IDEA Complete the key idea: ATP is the energy currency: …	ATP is the energy currency: ATP → ADP + P releases energy

KEY IDEA Complete the key idea: Photosynthesis: CO_2 + H_2O …	Photosynthesis: CO_2 + H_2O + light → glucose + O_2
KEY IDEA Complete the key idea: Respiration: glucose + O_2 → …	Respiration: glucose + O_2 → CO_2 + H_2O + about 30 ATP
KEY IDEA Complete the key idea: Anaerobic: only 2 ATP; …	Anaerobic: only 2 ATP; lactate in muscle, ethanol in yeast
QUICK CHECK Photosynthesis takes place in	chloroplasts
QUICK CHECK Respiration releases	carbon dioxide
QUICK CHECK Yeast fermentation produces	ethanol and CO_2
QUICK CHECK ATP → ADP releases	energy
PROBLEM Which gas is released by photosynthesis?	oxygen. Water is split, releasing O_2.
PROBLEM What is the cell's main energy currency?	ATP. Adenosine triphosphate.
PROBLEM How many molecules of O_2 are needed to respire one molecule of glucose?	6. From the balanced equation.
PROBLEM Glucose has M = 180 g/mol. How many moles of CO_2 are produced when 90 g of glucose is fully respired?	3 mol. × 6 = 3 mol CO_2.
PROBLEM Aerobic respiration gives about 30 ATP per glucose; anaerobic gives 2. How many times more ATP does aerobic give?	15 ×. 30 ÷ 2.

PROBLEM During a sprint, muscle cells short of oxygen produce	lactate. Anaerobic respiration in animals.
PROBLEM Why do leaves look green?	chlorophyll reflects green light while absorbing red and blue. Green is the colour least absorbed.
PROBLEM Explain why oxygen is essential for producing most of a cell's ATP.	In aerobic respiration, electrons from glucose pass along the electron transport chain in the mitochondria, releasing energy used to make most of the ATP. Oxygen is the final electron acceptor at the end of the chain, combining with electrons and hydrogen to form water. Without oxygen the chain backs up and stops, leaving only the 2 ATP from glycolysis.

Module 6.4 · DNA, genes and protein synthesis

EQUATION Write the equation: Chargaff's rule (from base pairing)	$[A] = [T], \qquad [G] = [C]$
EQUATION Write the equation: Central dogma	$\mathrm{DNA} \rightarrow \mathrm{RNA} \rightarrow \mathrm{protein}$
EQUATION Write the equation: Number of codons	$4^3 = 64$
KEY IDEA Complete the key idea: DNA: double helix, …	DNA: double helix, A–T and G–C pairing
KEY IDEA Complete the key idea: Replication is …	Replication is semi-conservative
KEY IDEA Complete the key idea: DNA → mRNA …	DNA → mRNA (transcription) → protein (translation)
KEY IDEA Complete the key idea: Codons are triplets; the …	Codons are triplets; the code is universal and redundant
QUICK CHECK A codon is	three bases

QUICK CHECK Translation happens at	ribosomes
QUICK CHECK RNA uses which base instead of thymine?	uracil
QUICK CHECK Whose X-ray images were crucial to the DNA structure?	Rosalind Franklin
PROBLEM Which base pairs with guanine in DNA?	cytosine. G–C.
PROBLEM Write the complementary strand to 5′-ATGCCT-3′ (reading the partner 3′ to 5′).	TACGGA. A→T, T→A, G→C, C→G.
PROBLEM A DNA sample is 30% adenine. What percentage is guanine?	20 %. G + C = 40%, G = 20%.
PROBLEM How many amino acids are coded by an mRNA sequence of 300 bases (ignoring stop codons)?	100. 3 bases per codon: 300 ÷ 3.
PROBLEM In transcription, thymine in DNA is matched in RNA by	adenine. The RNA base opposite a DNA T is A. (RNA uses U in place of T, opposite DNA's A.)
PROBLEM Why can a mutation have no effect on the protein?	the new codon may code for the same amino acid. The code is redundant.
PROBLEM Semi-conservative replication means each new DNA molecule has	one old and one new strand. Confirmed by Meselson and Stahl in 1958.
PROBLEM Why is the near-universal genetic code evidence for common ancestry?	The assignment of codons to amino acids is largely arbitrary chemically; many other codes would work. Yet bacteria, plants, fungi and animals almost all use the same one. The simplest explanation is that all life inherited it from a common ancestor, just as shared quirks in languages point to a common origin.

Module 6.5 · Inheritance and genetics

EQUATION Write the equation: Monohybrid cross ratios	$Aa \times Aa: \quad 1\,AA : 2\,Aa : 1\,aa$
EQUATION Write the equation: Chromosome combinations from independent assortment	$2^{23} \approx 8.4 \times 10^{6}$
EQUATION Write the equation: Hardy–Weinberg (allele frequencies p, q)	$p^2 + 2pq + q^2 = 1$
KEY IDEA Complete the key idea: Genotype: alleles you …	Genotype: alleles you carry. Phenotype: the trait shown
KEY IDEA Complete the key idea: Aa × Aa …	Aa × Aa gives 3:1 dominant:recessive phenotypes
KEY IDEA Complete the key idea: X-linked recessive traits …	X-linked recessive traits are more common in males
KEY IDEA Complete the key idea: Meiosis halves chromosome …	Meiosis halves chromosome number and creates variation
QUICK CHECK An organism with two different alleles is	heterozygous
QUICK CHECK Aa × Aa gives a phenotype ratio of	3:1
QUICK CHECK Meiosis produces cells with	half the chromosomes
QUICK CHECK X-linked recessive traits are commoner in	males
PROBLEM Two Aa parents. What fraction of offspring are expected to be aa? Give a decimal.	0.25. One square in four.

PROBLEM AA × aa. What fraction of offspring are heterozygous? Give a decimal.	1. Every offspring gets A from one parent and a from the other.
PROBLEM Aa × aa. What fraction show the dominant phenotype? Give a decimal.	0.5. Offspring: Aa, Aa, aa, aa.
PROBLEM Two carriers of cystic fibrosis (recessive) have three children. What is the probability none are affected? Give a decimal.	0.4219. $(3/4)^3 \approx 0.42$.
PROBLEM A woman who is a carrier of colour blindness (X^BX^b) has children with a man with normal vision. What fraction of their sons are expected to be colour blind?	1/2. Sons get Y from father and one X from mother: half get X^b.
PROBLEM A recessive allele has frequency q = 0.1. Using Hardy–Weinberg, what fraction of the population is homozygous recessive? Give a decimal.	0.01. $q^2 = 0.01$.
PROBLEM Why do siblings with the same parents look different?	Each parent's sex cells are made by meiosis. Independent assortment of 23 chromosome pairs gives over 8 million combinations, and crossing over shuffles alleles within chromosomes. Each child comes from a different egg and sperm, so siblings (except identical twins) inherit different combinations of their parents' alleles.

Module 6.6 · Evolution and natural selection

EQUATION Write the equation: Allele frequency change under selection	$p' = \frac{p(1+s)}{p(1+s)+(1-p)}$
EQUATION Write the equation: Relative fitness	$w = \frac{\text{offspring of a type}}{\text{offspring of the best type}}$
KEY IDEA Complete the key idea: Natural selection needs …	Natural selection needs variation, inheritance and differential reproduction

KEY IDEA Complete the key idea: Evolution = change in …	Evolution = change in allele frequencies in a population
KEY IDEA Complete the key idea: Evidence: fossils, …	Evidence: fossils, anatomy, DNA, observed change
KEY IDEA Complete the key idea: Drift is …	Drift is chance; speciation follows separation
QUICK CHECK Evolution happens to	populations
QUICK CHECK Darwin and Wallace published in	1858
QUICK CHECK Fitness in biology means	reproductive success
QUICK CHECK A scientific 'theory' is	a well-tested explanation
PROBLEM Which three things are needed for natural selection?	variation, inheritance, differential reproduction. Darwin's core logic.
PROBLEM Antibiotic resistance in bacteria is an example of	natural selection. Resistant variants survive and reproduce.
PROBLEM An allele has frequency 0.5 and a 10% advantage (s = 0.1). Using $p' = p(1+s)/[p(1+s) + (1 - p)]$, what is its frequency next generation?	0.5238. 0.55 ÷ (0.55 + 0.5) ≈ 0.524.
PROBLEM A human arm, a bat's wing and a whale's flipper share the same bone pattern. These are	homologous structures. Same structure from a common ancestor.
PROBLEM Genetic drift has the biggest effect in	small populations. Chance fluctuations matter more when numbers are small.
PROBLEM A giraffe stretching its neck to reach leaves will	not change its offspring's neck length. Acquired characteristics are not inherited.

PROBLEM What usually starts the formation of a new species?	separation of populations so they diverge. Isolation then divergence.
PROBLEM Explain why finishing a course of antibiotics matters, using natural selection.	A bacterial population varies in susceptibility. Antibiotics kill the most susceptible first. Stopping early leaves the hardiest survivors, which multiply and pass on their resistance genes, so the next infection is harder to treat. A full course aims to kill the whole population before resistant variants take over.

Module 6.7 · Human body systems

EQUATION Write the equation: Cardiac output	$\mathrm{CO} = \mathrm{HR} \times \mathrm{SV}$
EQUATION Write the equation: Body mass index (a population screening tool)	$\mathrm{BMI} = \frac{m\,(\mathrm{kg})}{h^{2}\,(\mathrm{m}^{2})}$
KEY IDEA Complete the key idea: Cells → tissues …	Cells → tissues → organs → systems
KEY IDEA Complete the key idea: Heart: double pump. Lungs: …	Heart: double pump. Lungs: alveoli give huge surface area
KEY IDEA Complete the key idea: Nerves: fast, electrical. …	Nerves: fast, electrical. Hormones: slower, in the blood
KEY IDEA Complete the key idea: Homeostasis by negative …	Homeostasis by negative feedback, e.g. insulin and glucagon
QUICK CHECK The organ that releases insulin is the	pancreas
QUICK CHECK Homeostasis mostly uses	negative feedback
QUICK CHECK Alveoli are adapted by having	a large surface area

QUICK CHECK The heart is best described as	a double pump
PROBLEM A resting heart beats 70 times a minute with a stroke volume of 70 mL. What is the cardiac output in litres per minute?	4.9 L/min. 70 × 70 = 4900 mL/min.
PROBLEM Where in the lungs does gas exchange occur?	alveoli. Thin-walled air sacs.
PROBLEM Which hormone lowers blood glucose?	insulin. Insulin, from the pancreas.
PROBLEM Sweating when hot is an example of	negative feedback. The response counters the change.
PROBLEM During exercise, heart rate rises to 150 beats/min and stroke volume to 110 mL. What is cardiac output, in L/min?	16.5 L/min. 150 × 110 = 16 500 mL/min.
PROBLEM A nerve impulse travels at 80 m/s. How long does it take to travel 1.6 m from toe to brain, in milliseconds?	20 ms. t = 1.6/80 = 0.02 s.
PROBLEM Compared with nervous signals, hormonal signals are generally	slower and longer-lasting. They travel in the blood.
PROBLEM Explain how the body corrects a fall in core temperature.	Temperature receptors detect the drop and the hypothalamus triggers responses: blood vessels in the skin narrow (vasoconstriction) so less heat is lost; muscles shiver, releasing heat from respiration; hair erects slightly. As temperature returns to normal, the responses switch off: negative feedback.

Module 6.8 · Microbes, immunity and disease

EQUATION Write the equation: Bacterial growth	$N = N_0 \times 2^{t/t_d}$
EQUATION Write the equation: Herd immunity threshold	$H = 1 - \frac{1}{R_0}$
EQUATION Write the equation: Effective reproduction number	$R_{\text{eff}} = R_0 \times (\text{fraction susceptible})$
KEY IDEA Complete the key idea: Bacteria are cells; …	Bacteria are cells; viruses need host cells
KEY IDEA Complete the key idea: Innate immunity: fast, general. …	Innate immunity: fast, general. Adaptive: slow, specific, with memory
KEY IDEA Complete the key idea: Vaccines train immune …	Vaccines train immune memory without the disease
KEY IDEA Complete the key idea: Herd immunity threshold = 1 …	Herd immunity threshold = 1 − 1/R₀; antibiotics don't work on viruses
QUICK CHECK Viruses reproduce	only inside host cells
QUICK CHECK Immune memory means the second response is	faster and stronger
QUICK CHECK If R_eff < 1, an epidemic	declines
QUICK CHECK Germ theory was established by	Pasteur and Koch
PROBLEM A bacterium divides every 20 minutes. Starting from one cell, how many are there after 2 hours?	64. $2^6 = 64$.

PROBLEM Antibiotics are effective against	bacteria. They target bacterial structures.
PROBLEM Measles has $R_0 \approx 15$. What is the herd immunity threshold, as a percentage?	93.33 %. About 93%.
PROBLEM A disease has $R_0 = 3$. What percentage must be immune for herd immunity?	66.67 %. 1 − 1/3 ≈ 67%.
PROBLEM With $R_0 = 4$ and 60% of people immune, what is R_eff?	1.6. 4 × 0.4 = 1.6: still spreading.
PROBLEM Vaccines work by	creating immune memory without causing the disease. They train the adaptive immune system.
PROBLEM Antibodies are made by	B cells. B lymphocytes.
PROBLEM Why can a person who is not vaccinated still be protected by others' vaccination?	If enough of the population is immune, each infected person meets too few susceptible people to pass the infection on to more than one other on average (R_eff below 1). Chains of transmission die out before reaching the unvaccinated, who are shielded indirectly. This is herd immunity.

Module 6.9 · Ecology and ecosystems

EQUATION Write the equation: Ecological efficiency	$E_{n+1} \approx 0.1\, E_n$
EQUATION Write the equation: Logistic growth	$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$
EQUATION Write the equation: Exponential growth	$N = N_0 e^{rt}$
KEY IDEA Complete the key idea: About 10% of energy …	About 10% of energy passes up each trophic level

KEY IDEA Complete the key idea: Energy flows; …	Energy flows; matter cycles
KEY IDEA Complete the key idea: Logistic growth levels …	Logistic growth levels off at carrying capacity K
KEY IDEA Complete the key idea: Human activity is reshaping …	Human activity is reshaping the carbon cycle and biodiversity
QUICK CHECK Energy in ecosystems	flows one way
QUICK CHECK Carrying capacity is	the maximum sustainable population
QUICK CHECK Decomposers	break down dead matter
QUICK CHECK Pre-industrial CO_2 was about	280 ppm
PROBLEM Producers in an ecosystem capture 20 000 kJ. About how much energy reaches the carnivores that eat herbivores?	200 kJ. Two steps at 10%: 20 000 → 2000 → 200.
PROBLEM Which process removes CO_2 from the atmosphere?	photosynthesis. Plants fix carbon.
PROBLEM Why are there fewer top predators than herbivores?	energy is lost at each trophic level. Only about 10% passes up each level.
PROBLEM In the logistic model, what is dN/dt when $N = K/2$, $r = 0.4$ per year and $K = 1000$?	100 per year. = 100 per year (the fastest growth).
PROBLEM And what is dN/dt when $N = K$?	0 per year. $1 - N/K = 0$, so growth stops.

PROBLEM Atmospheric CO_2 rose from 280 ppm to 420 ppm. What percentage increase is that?	50 %. (420 − 280)/280 = 0.5.
PROBLEM In predator–prey cycles, predator population peaks typically	follow prey peaks. Predators respond to food supply with a lag.
PROBLEM Explain why eating lower on the food chain can feed more people from the same land.	Only about 10% of the energy in plants ends up in the animals that eat them. Feeding crops directly to people uses that energy at the first step, while feeding them to livestock loses about 90% before it reaches people as meat. So the same crop area supports roughly ten times as many people eating plants as eating meat from those plants.